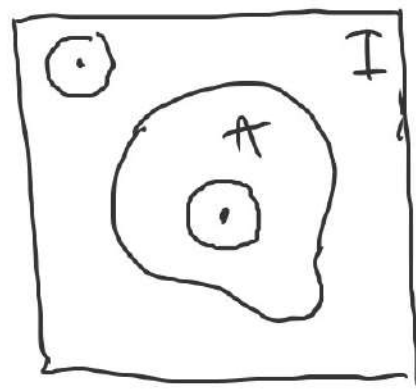


Fie $A \subset \mathbb{R}^n$ mărginită și $I \subset \mathbb{R}^n$ interval închis
a.î $A \subset I$.



$$\chi_A: \mathbb{R}^n \rightarrow \mathbb{R}, \quad \chi_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A. \end{cases}$$

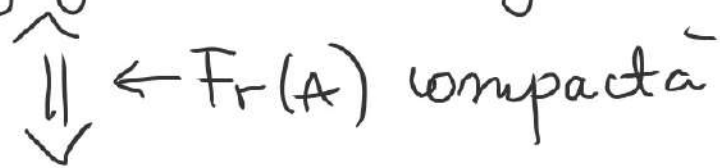
Propoziție Funcția χ_A este integrabilă Riemann pe A
dacă și numai dacă $A \in \mathcal{J}(\mathbb{R}^n)$

Dem. Disc $\chi_A = \text{Fr}(A)$

Gut. Lebesgue } χ_A and Riemann pe I



$F_r(A)$ neglijabilă Lebesgue



$\leftarrow F_r(A)$ compactă

$F_r(A) \in \mathcal{J}(\mathbb{R}^n)$ și $\lambda(F_r(A)) = 0$



$A \in \mathcal{J}(\mathbb{R}^n)$

Propoziție. Cu același notatî, avem.

$$\lambda^*(A) = \overline{\int_I \chi_A}, \quad \lambda_*(A) = \underline{\int_I \chi_A}$$

Demonstrație (exercițiu)

Definiție Fie $A \subset \mathbb{R}^n$ măsurabilă Jordan, $f: A \rightarrow \mathbb{R}$ mărg.
Fie $I \subset \mathbb{R}^n$ interval închis a.î. $A \subset I$. Definim

$$\tilde{f}: I \rightarrow \mathbb{R}, \quad \tilde{f}(x) = \begin{cases} f(x), & x \in A \\ 0, & x \in I \setminus A. \end{cases}$$

Spunem că f este integrabilă Riemann pe A dacă

dacă $\tilde{f}$ este integrabilă Riemann pe I .

$$\int_A f(x) dx := \int_I \tilde{f}(x) dx$$

Definiția nu depinde de alegerea intervalului I .

Teoremă Fie $A \subset \mathbb{R}^n$, $A \in \mathcal{J}(\mathbb{R}^n)$ și $f: A \rightarrow [0, \infty)$

integrabilă Riemann. Atunci

$$\Gamma_f = \left\{ (x, t) \in \mathbb{R}^n \times \mathbb{R} \mid x \in A, 0 \leq t \leq f(x) \right\} \in \mathcal{J}(\mathbb{R}^{n+1})$$

$$\text{și} \quad \lambda(\Gamma_f) = \int_A f(x) dx$$

Demonstratie. Exercitiu!

Corolar. Fie $A \in \mathcal{J}(\mathbb{R}^n)$ și $f: A \rightarrow \mathbb{R}$ integrabilă Riemann.

$$G_f = \{(x, f(x)) \mid x \in A\} \in \mathcal{J}(\mathbb{R}^n)$$

și $\lambda(G_f) = 0$. (pt că $G_f \subset \text{Fr}(\Gamma_f)$).

Corolar. Fie $A \in \mathcal{J}(\mathbb{R}^n)$, și $g: A \rightarrow \mathbb{R}$ integrabilă R.

a. i. $f \leq g$. Atunci

$$\Gamma_{f,g} = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid x \in A, f(x) \leq t \leq g(x)\} \in \mathcal{J}(\mathbb{R}^{n+1})$$

și

$$\lambda(\Gamma_{f,g}) = \int_A (g(x) - f(x)) dx.$$

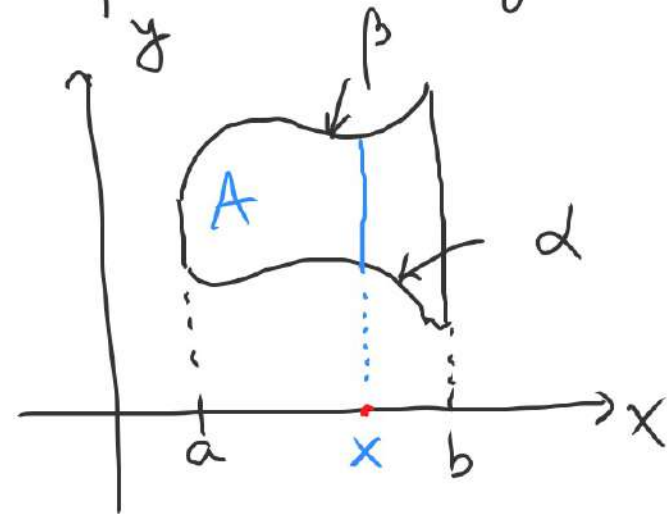
Propoziție 1. Fie $\alpha, \beta: [a, b] \rightarrow \mathbb{R}$ continue a.i. $\alpha(x) \leq \beta(x)$,
 $\forall x \in [a, b]$. Atunci

$$A = \{ (x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, \alpha(x) \leq y \leq \beta(x) \} \in \mathcal{J}(\mathbb{R}^2).$$

Dacă $f: A \rightarrow \mathbb{R}$ este continuă, atunci f este integr.

Riemann în

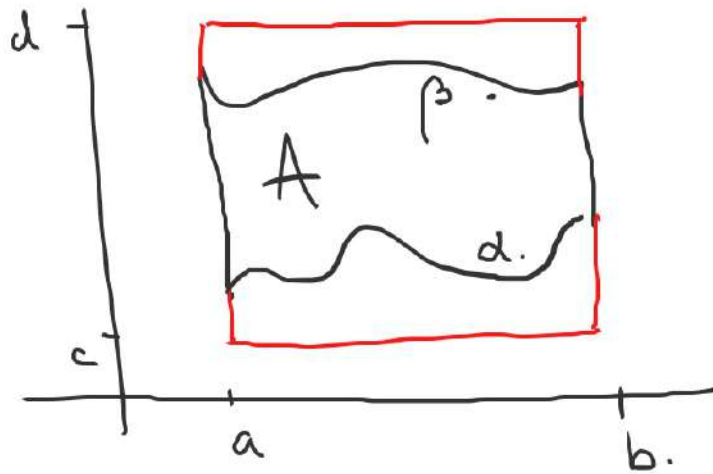
$$\iint_A f(x, y) dx dy = \int_a^b \left(\int_{\alpha(x)}^{\beta(x)} f(x, y) dy \right) dx$$



Demonstratie

Alegem $K = [c, d]$

ai. $A \subset [a, b] \times [c, d]$.



$A \in \mathcal{J}(\mathbb{R}^2)$ pt ca α si β fund cont pe $[a, b]$ sunt intez $\mathbb{R}$.

$$\tilde{f} : [a, b] \times [c, d] \rightarrow \mathbb{R} \quad \tilde{f}(x, y) = \begin{cases} f(x, y) & , (x, y) \in A \\ 0 & (x, y) \notin A \end{cases}$$

Desc. $\tilde{f}$ - multime neglijabila Lebesgue
(chiar neglij Jordan).

A compacta $\stackrel{f \text{ cont}}{\Rightarrow} f$ marg $\Rightarrow \tilde{f}$ marginte.

Out Lebesgue $\Rightarrow \tilde{f}$ integr. R pe $[a,b] \times [c,d] \Rightarrow f$ cont R pe A .

$$\iint_A f(x,y) dx dy = \iint_{[a,b] \times [c,d]} \tilde{f}(x,y) dx dy.$$

$$x \in [a,b], A_x = [\alpha(x), \beta(x)]$$

$$f_x: A_x \rightarrow \mathbb{R} \quad \tilde{f}_x: [c,d] \rightarrow \mathbb{R}$$

$$f_x(y) = f(x,y), \quad \tilde{f}_x(y) = \tilde{f}(x,y) = \begin{cases} f(x,y), & y \in A_x \\ 0 & , y \in [c,d] \setminus A_x \end{cases}$$

f_x cont pe $A_x \Rightarrow f_x$ est integrabile Riemann

$\Downarrow$

$\tilde{f}_x$ integr. Riemann,

Deci $\forall x \in [a, b]$, $\tilde{f}_x$ este integr. Riemann n.

$$\int_c^d \tilde{f}_x = \int_{d(x)}^{\beta(x)} f_x$$

T. Fubini $\Rightarrow x \mapsto \int_c^d \tilde{f}_x$ integr. Riemann pe $[a, b]$.

$$\int_a^b \left(\int_c^d \tilde{f}_x(y) dy \right) dx = \iint_{[a,b] \times [c,d]} \tilde{f}(x,y) dx dy = \iint_A f(x,y) dx dy$$

$$\int_a^b \left(\int_{d(x)}^{\beta(x)} f(x,y) dy \right) dx.$$

Calculati: $\iint_D (x+y) dx dy$, D este marginit de dreptele
 $y=x$, $y=0$ si $x=1$

$$D = \{(x,y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 \leq y \leq x\}$$

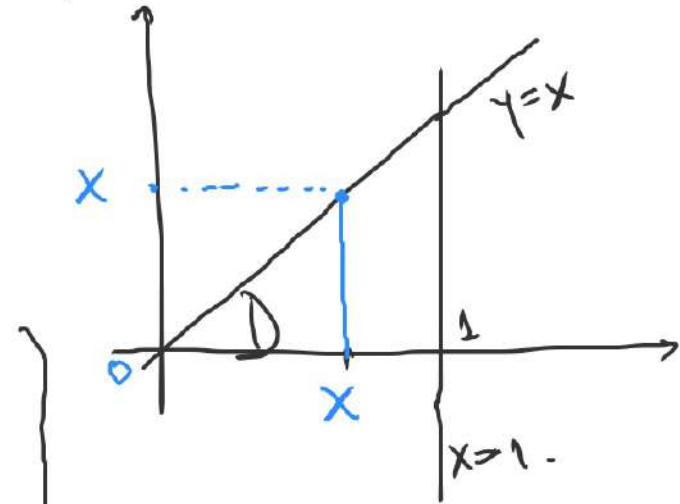
$$\alpha, \beta: [0,1] \rightarrow \mathbb{R}$$

$$\alpha(x) = 0, \beta(x) = x.$$

α, β integr. R pt că sunt cont pe $[a,b]$

f cont pe $D \xrightarrow{\text{Prop 1}} f$ integrabilă Riemann si

$$f(x,y) = x+y.$$



$$\Rightarrow D = T_{\alpha, \beta} \in \mathcal{J}(\mathbb{R}^2).$$

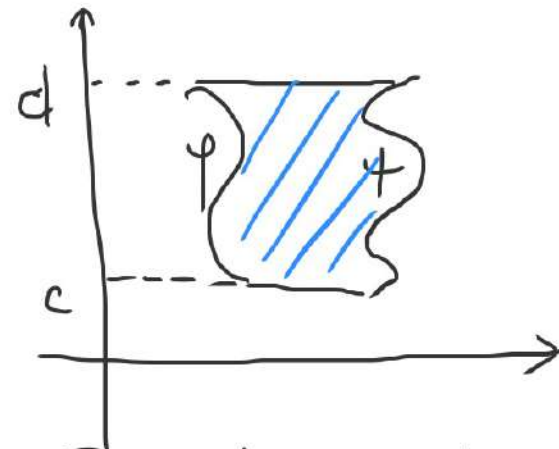
$$\iint_D (x+y) dx dy = \int_0^1 \left(\int_0^x (x+y) dy \right) dx$$

$$= \int_0^1 \left(x \int_0^x dy + \int_0^x y dy \right) dx = \int_0^1 \left(x \cdot y \Big|_{y=0}^{y=x} + \frac{y^2}{2} \Big|_{y=0}^{y=x} \right) dx$$

$$= \int_0^1 \left(x^2 + \frac{x^2}{2} \right) dx = \frac{3}{2} \cdot \frac{x^3}{3} \Big|_0^1 = \frac{1}{2}$$

Propozitie 2. Fie $\varphi, \psi: [c, d] \rightarrow \mathbb{R}$ continue a.i.

$\varphi(y) \leq \psi(y)$, $\forall y \in [c, d]$ Atunci



$$A = \{(x, y) \in \mathbb{R}^2 \mid c \leq y \leq d, \varphi(y) \leq x \leq \psi(y)\}$$

este masurabilă Jordan. Dacă $f: A \rightarrow \mathbb{R}$ este continuă
at. f este int. $\mathbb{R}$ pe A și

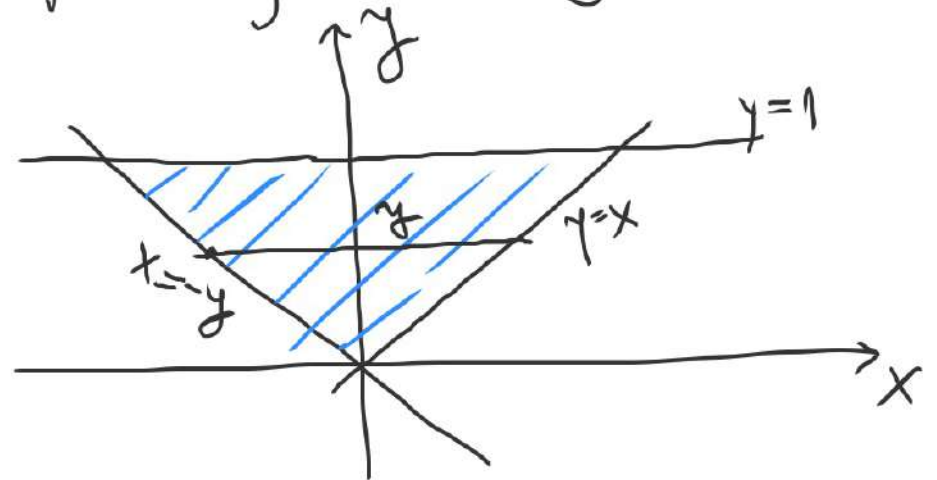
$$\iint_A f(x, y) dx dy = \int_c^d \left(\int_{\varphi(y)}^{\psi(y)} f(x, y) dx \right) dy$$

Calculati $\iint_A (x+xy) dx dy$ unde

A este marginita de dreptele $y=x$, $y=-x$ si $y=1$.

Solutie.

$$A = \{(x,y) \in \mathbb{R}^2 \mid 0 \leq y \leq 1, \underbrace{-y \leq x \leq y}_{\text{cont.}}\}$$



$$A \in \mathcal{J}(\mathbb{R}^2)$$

$f: A \rightarrow \mathbb{R}$, $f(x,y) = x+xy$ este continua pe A .

Prop 2 $\Rightarrow f \in \text{int. } \mathbb{R}$ si

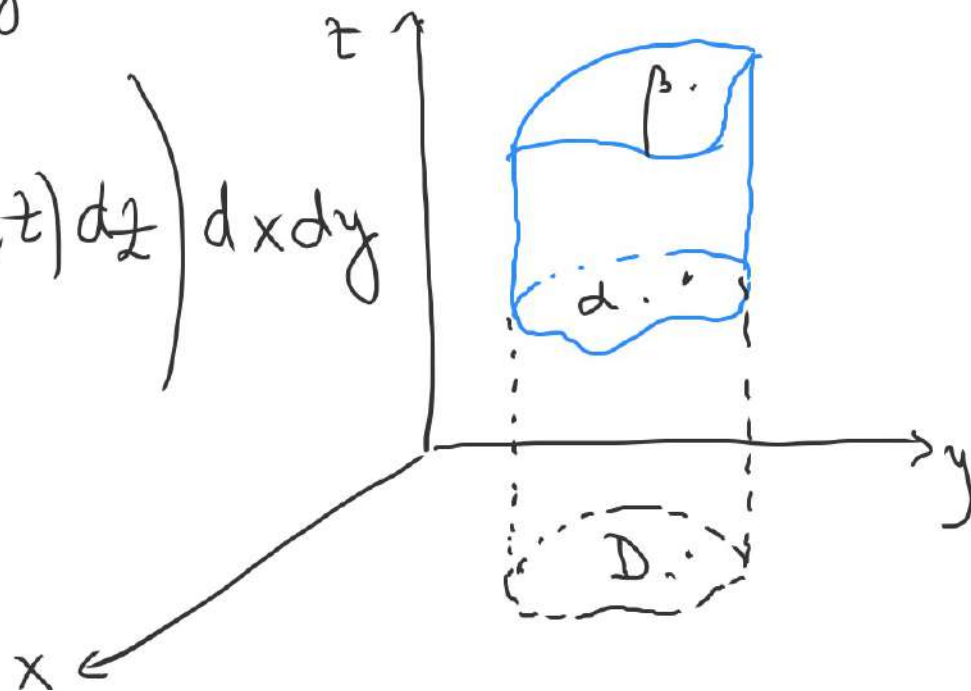
$$\begin{aligned}\iint_A (x+xy) dx dy &= \int_0^1 \left(\int_{-y}^y (x+xy) dx \right) dy \\ &= \int_0^1 \left(\frac{x^2}{2} + y \cdot \frac{x^2}{2} \right) \bigg|_{x=-y}^{x=y} dy = \int_0^1 0 dy = 0.\end{aligned}$$

Propoziție 3 Fie $D \in \mathcal{J}(\mathbb{R}^2)$, $\alpha, \beta: D \rightarrow \mathbb{R}$ continue și marginale pe D a.i. $\alpha(x, y) \leq \beta(x, y)$, $\forall (x, y) \in D$ Atunci

$$V = \{ (x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D, \alpha(x, y) \leq z \leq \beta(x, y) \}$$

este masurabilă Jordan. Dacă $f: V \rightarrow \mathbb{R}$ continuă și marg pe V at. f este integrabilă Riemann și

$$\iiint_V f(x, y, z) dx dy dz = \iint_D \left(\int_{\alpha(x, y)}^{\beta(x, y)} f(x, y, z) dz \right) dx dy$$



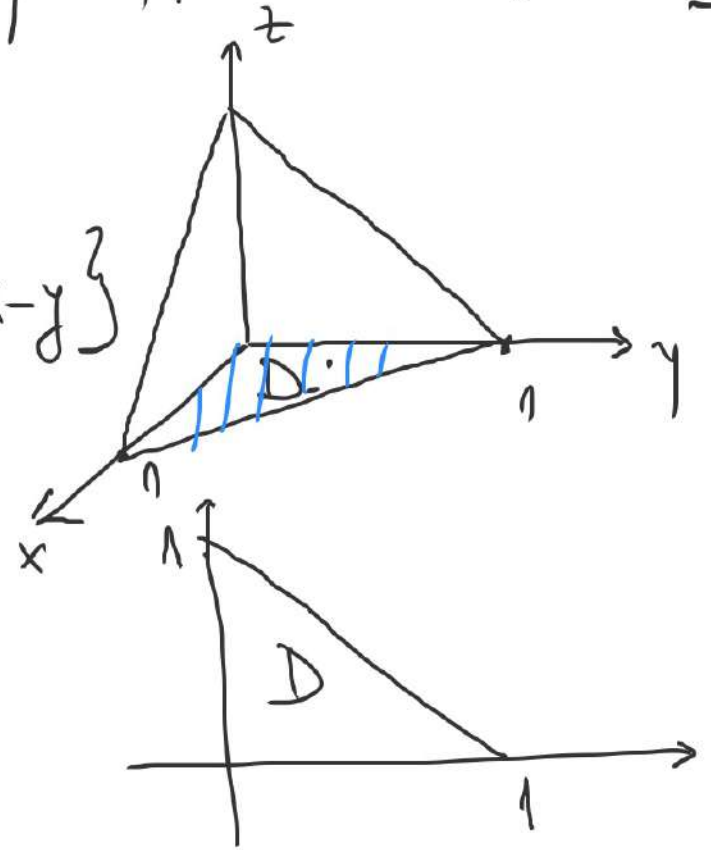
Calculati $\iiint_V x dx dy dz$, $V = \{(x, y, z) \mid x+y+z \leq 1, x, y, z \geq 0\}$

$$V = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D, 0 \leq z \leq 1-x-y\}$$

$$D = \{(x, y) \in \mathbb{R}^2 \mid x+y \leq 1, x, y \geq 0\}$$

$$\parallel$$

$$\{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 \leq y \leq 1-x\}$$



$$D \in \mathcal{J}(\mathbb{R}^2), V \in \mathcal{J}(\mathbb{R}^3) \text{ si } f(x, y, z) = x \text{ int. } \mathbb{R}. \text{ (exercitiu).}$$

↑
Prop 1.

↑
Prop 3.

$$\iiint_V x \, dx \, dy \, dz = \iint_D \left(\int_0^{1-x-y} x \, dz \right) dx \, dy$$

$$= \iint_D \left(x \cdot z \Big|_{z=0}^{z=1-x-y} \right) dx \, dy = \iint_D x(1-x-y) \, dx \, dy =$$

$$= \int_0^1 \left(\int_0^{1-x} (x - x^2 - xy) \, dy \right) dx$$

$$= \int_0^1 \left(xy - x^2 y - \frac{xy^2}{2} \right) \Big|_{y=0}^{y=1-x} dx = \int_0^1 \left[x(1-x) - x^2(1-x) - \frac{x(1-x)^2}{2} \right] dx = \dots$$

